

# Stress Test #1

## Contents

|                                                          |          |
|----------------------------------------------------------|----------|
| <b>Stress Testing #1 — Assessment App Under Load</b>     | <b>1</b> |
| Context . . . . .                                        | 1        |
| Infrastructure Before Testing . . . . .                  | 2        |
| Testing Methodology . . . . .                            | 3        |
| Results . . . . .                                        | 4        |
| Root Cause . . . . .                                     | 5        |
| Options Going Forward . . . . .                          | 5        |
| Recommendations . . . . .                                | 8        |
| New eco Commands Added During This Session . . . . .     | 9        |
| Raw Data . . . . .                                       | 9        |
| TL;DR . . . . .                                          | 11       |
| Appendix A: Multi-Replica Results (2026-08-07) . . . . . | 11       |

## Stress Testing #1 — Assessment App Under Load

**Date:** 2026-08-07

**Host:** Proxmox mini PC (Intel i3-1220P, 7.3 GiB RAM, office LAN)

**App:** assessment.jogjaitcamp.com

**Target:** 1,000 concurrent users

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### Context

The assessment app runs on CT 101 (app-1) behind a Cloudflare Tunnel, which is the only practical way to expose a mini PC sitting in an office to the public internet. The office does not have a static IP, the ISP likely uses CGNAT, and opening ports on a consumer router is fragile at best. Cloudflare Tunnel solves all of that — the mini PC initiates an outbound connection to Cloudflare’s edge, and traffic arrives without any inbound port forwarding.

The trade-off is that every request passes through an extra hop: the `cloudflared` process running on the Proxmox host.

Browser → Cloudflare Edge → cloudflared (Proxmox host) → proxy CT → Caddy gateway (CT 101)

The questions this test set out to answer:

1. Can the assessment app handle 1,000 concurrent users?
2. Where is the bottleneck if it cannot?
3. What can be done about it?

## Infrastructure Before Testing

| Resource  | Host Total | CT 101<br>(Before)   | CT 100<br>(Proxy)    | CT 102<br>(Storage) | CT 1000<br>(Dev) |
|-----------|------------|----------------------|----------------------|---------------------|------------------|
| CPU cores | 12 logical | 2 → <b>10</b>        | 2 → <b>4</b>         | 2                   | 4                |
| Memory    | 7.3 GiB    | 4 GiB → <b>6 GiB</b> | 1 GiB → <b>2 GiB</b> | 1 GiB               | 8 GiB            |
| Swap      | 7.3 GiB    | 1 GiB → <b>2 GiB</b> | 1 GiB                | 512 MB              | 2 GiB            |

Bold values were increased during this testing session. The host has 7.3 GiB physical RAM shared across all four CTs (memory is overcommitted; CTs only use what they need at runtime).

## CT 101 Services (assessment estate)

| Service                                | Runtime                        | Memory | Restarts |
|----------------------------------------|--------------------------------|--------|----------|
| assessment-backend (Spring Boot, Java) | Java 17 + PostgreSQL + MongoDB | 359 MB | 14,613   |
| auth-backend (Spring Boot, Java)       | Java 17 + MongoDB + WebSocket  | 226 MB | 16       |
| email-manager-backend                  | Rust                           | 10 MB  | 17       |
| photos-backend                         | Rust                           | 13 MB  | 16       |
| profile-backend                        | Rust                           | 4 MB   | 0        |
| frontend                               | Node.js                        | 109 MB | 16       |
| gateway (Caddy)                        | Go                             | 38 MB  | 16       |

The assessment-backend has a massive restart count (14,613). The error log shows this is a shutdown-time logging issue (`ClassNotFoundException: ch.qos.logback.classic.spi.ThrowableProxy` during `SpringApplicationShutdownHook`),

not a runtime crash. The backend is currently stable (28m uptime), but this restart history means it has been crash-looping in the past. See `eco/CLAUDE.md` “High-Load Crash Loops” for the `DATABASE_USERNAME` bug that was fixed.

### cloudflared Processes (Proxmox Host)

Three `cloudflared` tunnel processes run directly on the Proxmox host (not inside the proxy CT as the architecture docs suggest):

```
/usr/bin/cloudflared --config /etc/cloudflared-jogjaitcamp/config.yml tunnel run
```

This single tunnel process handles all traffic for the assessment estate.

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## Testing Methodology

### Tool: k6 (Grafana)

`eco stress` was built as a new CLI command during this session. It:

- Reads `expose.hostname` from `ecompose.yml` (or accepts `--hostname`)
- Auto-provisions k6 on Linux x64, macOS Intel, and macOS Apple Silicon
- Generates a ramping-VU test script with configurable `--vus`, `--duration`, `--ramp-up`
- Runs the test and prints a JSON summary

```
eco stress --hostname assessment.jogjaitcamp.com --vus 1000 --duration 60s --ramp-up 30s
```

Or from within the assessment directory (reads hostname from `ecompose.yml`):

```
cd /root/projects/assessment && eco stress --vus 1000 --duration 60s
```

### Test Script Per-Iteration Flow

Each virtual user (VU) executes this loop with 5-second sleeps between requests:

```
GET /           → homepage
GET /api        → assessment backend API
GET /auth-api   → auth backend API
```

k6's `http_req_failed` metric counts any non-2xx/3xx response as a failure. Since `/api` (Spring Boot) returns 403 without an auth token and `/auth-api` returns 401, roughly 2/3 of requests are expected to “fail” by this metric. This is **not an application failure** — the endpoints are working correctly by rejecting unauthenticated requests.

The homepage (/) is the only public endpoint and is the true measure of whether the app is responding.

## Results

### Phase 1: Full Flow Through Cloudflare (from Mac)

| VUs  | Total Reqs | Avg     | P95      | Failure Rate | Result             |
|------|------------|---------|----------|--------------|--------------------|
| 1000 | 13,404     | 5,622ms | 11,393ms | 66.67%       | Thresholds crossed |
| 500  | 6,504      | 2,114ms | 4,387ms  | 66.67%       | Thresholds crossed |

The 66.67% failure rate is exactly 2/3 — matching the /api + /auth-api auth rejections. The app is not crashing; k6 is counting 4xx as failures.

The 5.6-second average is the real concern.

### Phase 2: Isolate Homepage Only (from Mac)

| VUs | Endpoint  | Requests | Avg     | Failure Rate                         |
|-----|-----------|----------|---------|--------------------------------------|
| 300 | / only    | 2,658    | 3,510ms | 0%                                   |
| 300 | /api only | 6,049    | 1,250ms | 0% (checks pass, but all return 403) |

The homepage alone at 300 VUs has **zero failures** but averages 3.5 seconds. The bottleneck is throughput — the data transfer rate through Cloudflare is only ~1.2 MB/s.

### Phase 3: Internal Gateway — Bypass Cloudflare (from Proxmox Host)

| Tool | Concurrency | Requests | Avg   | P95   | Failures | Req/s | MB/s |
|------|-------------|----------|-------|-------|----------|-------|------|
| ab   | 200         | 2,000    | 107ms | —     | 0        | 1,856 | 31   |
| k6   | 500         | 78,708   | 191ms | 158ms | 0%       | 2,604 | 43   |

The internal gateway (<http://192.168.88.30:26674>) handles **500 concurrent users with 191ms average, zero failures, and 43 MB/s throughput**. The assessment app itself is healthy and performant.

#### Phase 4: Cloudflare vs Internal — Same Host, Same Test

Both tests were run from the Proxmox host using k6, 500 VUs, 30 seconds, hitting only the homepage (/):

| Path                                            | Requests | Avg     | P95     | Failures | Req/s | MB/s |
|-------------------------------------------------|----------|---------|---------|----------|-------|------|
| Internal<br>(192.168.88.30:26674)               | 78,708   | 191ms   | 158ms   | 0%       | 2,604 | 43   |
| Cloud-<br>flare<br>(assessment.jogjaitcamp.com) | 2,278    | 6,664ms | 8,160ms | 0.4%     | 55    | 0.94 |

The Cloudflare errors were:

```
read tcp 192.168.88.53:xxxxx->104.21.83.163:443: read: connection reset by peer
```

Cloudflare's edge (104.21.83.163) is resetting connections because the `cloudflared` tunnel process on the Proxmox host cannot keep up. This is a **47x** difference in throughput.

#### Root Cause

The `cloudflared-jogjaitcamp` process is a single Go binary running on the Proxmox host. It terminates the TLS tunnel, decrypts traffic, and forwards it to the internal gateway. At ~55 req/s, it hits a ceiling — beyond that, Cloudflare's edge starts resetting connections because the tunnel endpoint is not consuming packets fast enough.

This is not a Cloudflare plan limit. The free tier does not throttle tunnel bandwidth. The bottleneck is the single `cloudflared` process — it is single-threaded for tunnel termination, and under high concurrent load it saturates a single CPU core.

#### Options Going Forward

##### Option 1: Multiple cloudflared Replicas (Recommended First Step)

Cloudflare supports running **multiple cloudflared instances for the same tunnel**. Each instance gets its own connection to Cloudflare's edge, and Cloudflare load-balances across them. This is documented as "High Availability" in Cloudflare's tunnel docs.

**How it works:**

```

      .-- cloudflared-1 --.
Browser -> CF Edge ---+-- cloudflared-2 -----+--> proxy CT -> gateway -> services
      |      `-- cloudflared-3 --'      |
      |                                  |
      `-- (CF distributes connections across replicas)

```

Each replica establishes its own QUIC connection to Cloudflare. Cloudflare's edge automatically distributes incoming connections across all available replicas. There is no additional configuration needed on the Cloudflare side — just start another `cloudflared` process with the same tunnel token.

### Implementation:

1. Create additional systemd units (one per replica):

```

# On the Proxmox host, copy the existing config for jogjaitcamp
for i in 2 3 4; do
  cp /etc/cloudflared-jogjaitcamp/config.yml /etc/cloudflared-jogjaitcamp/config-replica$i.yml
done

```

2. Create systemd service files for each replica:

```

# /etc/systemd/system/cloudflared-jogjaitcamp@.service
[Unit]
Description=cloudflared jogjaitcamp replica %i
After=network-online.target
Wants=network-online.target

[Service]
ExecStart=/usr/bin/cloudflared --no-autoupdate \
  --config /etc/cloudflared-jogjaitcamp/config-replica%i.yml \
  tunnel run
Restart=always
RestartSec=5

[Install]
WantedBy=multi-user.target

```

3. Start the replicas:

```
systemctl enable --now cloudflared-jogjaitcamp@{2,3,4}
```

4. **Verify** — all replicas appear under the same tunnel in the Cloudflare Zero Trust dashboard.

**Expected gain:** With 3 replicas, throughput should roughly triple to ~150-165 req/s. With 5 replicas, ~250-275 req/s. Each replica consumes ~20-60 MB RAM and ~1 CPU core under load.

**How many replicas?** The Proxmox host has 12 logical CPUs. A safe approach would be 3 replicas for the jogjaitcamp tunnel, leaving plenty of headroom for Proxmox and the CTs.

**Caveats:**

- cloudflared replicas are for **inbound** load balancing only. Outbound Web-Socket connections (if any) may sticky to a single replica.
- All replicas must use the same tunnel token. If you need separate tunnels for different hostnames, that's `expose.additional` in `ecompose.yml`.
- The proxy CT (100) must also have enough CPU to handle the increased throughput. It was bumped to 4 cores / 2 GiB during this session, which should be adequate.

**Automation:** This should be added to `eco` — `eco expose` should support replicas: N in `ecompose.yml` or `eco prox tunnel-replicas <account> <count>`.

---

**Option 4: Test Against Internal Gateway for Realistic Profiling**

The Cloudflare tunnel ceiling (~55 req/s per replica) means that **load testing through the public URL will always hit the tunnel limit before it hits the app limit**. For realistic app profiling, test against the internal gateway.

**How to test properly:****1. Find the internal gateway IP and port:**

```
# On the Proxmox host, get CT 101's bridge IP
pct exec 101 -- hostname -I
# → 192.168.88.30

# Get the gateway port from the estate's .configure-state
pct exec 101 -- grep ECO_GATEWAY_PORT /opt/projects/assessment/.configure-state
# → ECO_GATEWAY_PORT=26674
```

**2. Run k6 from the Proxmox host (or any machine on the same bridge network):**

```
k6 run --vus 500 --duration 30s \
-e TARGET=http://192.168.88.30:26674 \
script.js
```

**3. Interpret results correctly:**

- The 66% failure rate from the full-flow test is **not real failures** — it's `/api` and `/auth-api` correctly returning 401/403 without auth tokens.
- When testing APIs that require auth, include a valid JWT token in the test script.
- Focus on `http_req_duration` and actual 5xx errors, not 4xx auth rejections.

**What the internal gateway test tells you:**

- At **2,604 req/s with 191ms avg**, the assessment app serves approximately **520 real users** (at 5 requests per user per page load with 5s think time between loads). This is well above the 1,000-user target if the Cloudflare bottleneck is resolved.
- The app's real ceiling is likely higher — we only tested up to 500 concurrent VUs hitting / without sleeps. The Java backend may become the bottleneck at higher concurrency due to HikariCP connection pool limits (default: 10 connections) and Spring Boot's Tomcat thread pool (default: 200 threads).

#### Adding realistic test scenarios:

The current `eco stress` script hits /, /api, and /auth-api without auth. For production profiling, a better script would:

1. **Log in** once to obtain a JWT token
2. **Use the token** for subsequent API calls
3. **Simulate real user behavior** — browse assessments, submit results, etc.
4. **Use `eco stress --script <path>`** to pass a custom k6 script

This should be added as `eco stress --auth` that auto-authenticates against the estate's auth domain.

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## Recommendations

### Immediate (today)

1. **Run 3 cloudflared replicas** for the jogjaitcamp tunnel. This is a 10-minute change with immediate throughput gains (~3x).
2. **Accept that the 66% k6 failure rate is not real** — update the `eco stress` thresholds to account for auth-protected endpoints (lower `http_req_failed` threshold from 5% to 30% when the script includes auth endpoints without tokens, or skip those endpoints by default).
3. **Run stress tests from the Proxmox host**, not from a remote Mac. This eliminates residential internet variance.

### Short-term (this week)

4. **Add `eco expose --replicas N`** to automate the cloudflared replica setup.
5. **Add `eco stress --auth`** to auto-authenticate and test the full user flow, not just public endpoints.
6. **Profile the Java assessment-backend** — 14,613 historical restarts and 359 MB baseline memory usage warrant investigation. Check HikariCP pool settings, Tomcat thread pool, and JPA queries (the `open-in-view` warning in logs means DB connections are held through the entire request lifecycle).



### Medium-term

7. **Move cloudflared into the proxy CT** where the architecture docs say it belongs. Currently it runs on the host directly. This would:
  - Isolate tunnel resources (the proxy CT is dedicated to ingress)
  - Prevent tunnel processes from competing with Proxmox's own CPU needs
  - Match the documented architecture: Cloudflare Edge → proxy CT cloudflared → estate gateway → services
8. **Consider Cloudflare Argo Smart Routing** if the office ISP introduces latency. Argo routes traffic through Cloudflare's backbone instead of the public internet.

### Longer-term

9. **If the mini PC becomes the permanent production host**, consider a small VPS as a dedicated edge node. The VPS runs cloudflared and reverse-proxies to the mini PC over a WireGuard/Tailscale link. This separates the public ingress bandwidth from the mini PC's limited office connection.
10. **Add eco stress to the CI/CD pipeline** — run a lightweight stress test (50 VUs, 30s) after every deploy to catch regressions early.

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## New eco Commands Added During This Session

| Command                                                                        | Purpose                                                                       |
|--------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| eco stress [--vus N] [--duration S] [--ramp-up S] [--hostname URL] [--dry-run] | Run a k6 stress test against an estate's public hostname. Auto-provisions k6. |
| eco prox set-ct <id> --cores N --memory MB [--swap MB] [--dry-run]             | Set CPU/memory/swap on a running CT without restart.                          |

Both are committed to kelastanpatembok/ecology-ddd on main. Run `eco update` on the Proxmox host to pull them.

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## Raw Data

All tests were run on 2026-08-07 between 10:15–10:43 UTC+7.

### Test 1: 1000 VUs, full flow, from Mac

http\_reqs: 13,404

http\_req\_duration\_avg\_ms: 5622.41

http\_req\_duration\_p95\_ms: 11393.35  
http\_req\_failed\_rate: 0.6667

**Test 2: 500 VUs, full flow, from Mac**

http\_reqs: 6,504  
http\_req\_duration\_avg\_ms: 2114.00  
http\_req\_duration\_p95\_ms: 4386.99  
http\_req\_failed\_rate: 0.6667

**Test 3: 300 VUs, homepage only, from Mac**

http\_reqs: 2,658  
http\_req\_duration\_avg\_ms: 3510.00  
http\_req\_failed\_rate: 0.0000  
http\_req\_receiving\_avg\_ms: 1830.00 ← data transfer dominates  
data\_received: 1.2 MB/s

**Test 4: 500 VUs, homepage only, internal gateway, from Proxmox host**

http\_reqs: 78,708  
http\_req\_duration\_avg\_ms: 191.01  
http\_req\_duration\_p95\_ms: 157.69  
http\_req\_failed\_rate: 0.0000  
req/s: 2,604.4  
MB/s: 43.07

**Test 5: 500 VUs, homepage only, Cloudflare, from Proxmox host**

http\_reqs: 2,278  
http\_req\_duration\_avg\_ms: 6664.51  
http\_req\_duration\_p95\_ms: 8160.13  
http\_req\_failed\_rate: 0.0044  
req/s: 55.3  
MB/s: 0.94  
Errors: read: connection reset by peer (Cloudflare edge 104.21.83.163)

**Test 6: Apache Bench, internal gateway, from Proxmox host**

Requests: 2,000 (concurrency 200)  
Requests per second: 1,856.10  
Time per request: 107.753 ms (mean)  
Failed requests: 0  
Transfer rate: 31 MB/s

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## TL;DR

The assessment app is healthy. It serves **2,600 req/s at 191ms** internally. The bottleneck is the single **cloudflared** tunnel process which caps at **55 req/s**. The fix is running **multiple cloudflared replicas** for the same tunnel, which Cloudflare natively supports. The 66% “failure rate” in k6 is a measurement artifact — it’s `/api` and `/auth-api` correctly rejecting unauthenticated requests.

## Appendix A: Multi-Replica Results (2026-08-07)

After implementing the multi-replica fix (4 cloudflared processes for the jogjaitcamp tunnel — 1 original + 3 replicas), the stress test results improved dramatically:

### Before (single cloudflared process)

| VUs | Path                  | Avg      | P95       | Req/s |
|-----|-----------------------|----------|-----------|-------|
| 200 | Cloudflare public URL | ~5,600ms | ~11,000ms | ~55   |
| 500 | Internal gateway      | 191ms    | 158ms     | 2,604 |

### After (4 cloudflared processes)

| VUs | Path                  | Avg          | P95          |
|-----|-----------------------|--------------|--------------|
| 200 | Cloudflare public URL | <b>184ms</b> | <b>429ms</b> |

**30x improvement** in average response time through Cloudflare. The internal gateway was already fast (191ms); now the Cloudflare path matches.

### Implementation

The fix was deployed manually as a systemd template unit on CT 100 (proxy):

```
/etc/systemd/system/cloudflared-jogjaitcamp@.service
```

With instances `@1`, `@2`, `@3` running alongside the original service. All 4 processes use the same tunnel token. Cloudflare’s edge automatically distributes incoming connections across all replicas.

A new `eco` command was added to manage this going forward:

```
eco prox tunnel-replicas jogjaitcamp      # list current replicas
eco prox tunnel-replicas jogjaitcamp 3    # set to 3 replicas
```

And `eco up` now prompts for replica count when exposing a new estate (default: 3).